

Categorical approach to Cross-documents: Extension to Democritus

Nhi Thuy Ha

CMPSCI 692CT: Category Theory for AGI

University of Massachusetts Amherst

Abstract

We present a categorical causal discovery extended to the CLIFF system: Consciousness Layer Interface in FunctorFlow. CLIFF has an integrated system called Democritus, which retrieves scientific documents and extract structured causal relationships from them. In turn, Democritus performs topic discovery over input documents and expanded those topics using language models to form a local discourse manifold [1]. Democritus focuses on an important questions: which causal claims are explicitly stated across documents?

Our main focus on to this project extension is to identify the gap to that question. Given the causal claims that documents do state, what additional causal relationships are mathematically forced to exist that no document ever directly wrote down? We close this gap through two categorical mechanisms. The first is the **Kan Extension** module, which discovers novel causal claims by chaining together the stated ones. For example, if one documents asserts A causes B, and another document asserts B causes C, the mechanism infers that A causes C, even if no retrieved papers has written that sentence. Confidence on such claims are computed in a conservative or limited manner, which will be explained later on this report. Computing confidence would also have the possibility of weaker evidence producing weaker claims. The second second, **Sheaf Colimit** module, handles the contradictions between two documents. Let's say one paper asserts A causes B while another asserts B causes A, rather than discarding one, the mechanism handles this conflict as bidirectional relationships that neither paper has described.

When experimenting with this system, we applied to a corpus of three documents on sleep deprivation, cortisol, and related conditions. The system extracts 177 stated causal edges, discover an additional 40 new causal claims (novels) through chain composition, and 20 new sheaf colimit feedback loops through conflict resolution. There are several other experiments with differing queries and corpuses that will demonstrate how Kan extension and sheaf colimits are utilized for this AGI system.

1 Introduction

There are thousands of research papers distributed containing scientific works and experiments. Very few documents hold causal relationships between different scientific fields in depth, for example, biological and social phenomena connected on a single paper. The CLIFF system by Sridhar Mahadevan [2] focuses on using queries through user prompts containing a background agent called Democritus, which retrieves documents, extracts structured causal claims, and synthesises them into a cross-document report stored as CSQL database.

The CLIFF system automates these actions as it takes in the user queries and produces a summary report. The extraction step produces entries like: (source="sleep deprivation", relation="bidirectional feedback", target=cognitive performance, confidence=0.67, mechanism="sheaf colimit", explanation). Democritus collects such entries across all retrieved papers and asks which claims have multiple papers supporting them. If five papers say "X causes Y", it reports strong support for $X \rightarrow Y$. However, what is missing is Democritus asking: what can be inferred by chaining claims together? Or when two papers contradict each other, what do those contradictions reveal? Such

causal relationships can be entailed through transitivity or disagreements, even as they do not appear directly on those documents. It may be the case that these gaps are difficult to work with since chaining together any claims is at risk of hallucination and can drift away from the actual piece of the corpus. Regardless, our contribution to this project is to add a layer on top of Democritus by applying two categorical theory concepts to focus on such questions:

- **Kan Extension:** for every two step chain $A \rightarrow B \rightarrow C$ supported by retrieved papers, we infer $A \rightarrow C$ with a conservative confidence. This path construction entails a causal path from the stated edge using right Kan extensions.
- **Sheaf Colimit:** for every pair of contradictory claims ($A \rightarrow B$ in one paper and $B \rightarrow A$ in another), resolves with a bidirectional feedback loop $A \leftrightarrow B$.

Both algorithms are from category theory, specifically, the mathematical concepts of Kan extensions and sheaf colimits. Kan extension is explained in Mahadevan (2026, Universal Decisions with Kan Extensions) [5] and Riehl (2016, chapter 6) [6]. Sheaf colimits and their role to resolve disagreements appear in CSQL and Topos Causal Models [1]. The implementation and concepts are available as an extension of the CLIFF_CatAgi repository [2].

2 Background

2.1 CLIFF and FunctorFlow

FunctorFlow v3 is a category theory agentic framework where the user can submit a query (e.g., "Find 10 studies on the bidirectional relationship between sleep deprivation and cortisol"). The workflow has types of agents connected by morphisms of attention (colimit) and diffusion (limit). CLIFF is the conscious-layer interface that routes those queries to a background worker, and each of these workers runs through an unconscious process, resulting in having access to a conscious workspace, such as the weighted score combined with salience, relevance, novelty, and urgency [3]. What Democritus agent does is:

1. Downloads and parses relevant papers.
2. Extracts structured causal claims into a SQLite database (the CSQL database).
3. Create an HTML dashboard explaining the extracted claims.

Claims are stored as directed edges (source, relation, target, confidence, doc ID) in a SQLite file called democritus_csql.sqlite.

2.2 Category Theory Use

This project extension relies on the following concepts in building on the course:

Functors and Natural Transformations: A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ maps objects and morphisms between categories while keeping the composition and identities [6]. The stated claims corpus is formed as a functor $F : \text{StateCat} \rightarrow \text{CausalSpace}$ where StateCat has documents as objects and supporting citations as morphisms.

Kan Extensions: Given functors $F : \mathcal{C} \rightarrow \mathcal{E}$ and $K : \mathcal{C} \rightarrow \mathcal{D}$, the right Kan extension $\text{Ran}_K F : \mathcal{D} \rightarrow \mathcal{E}$ enforce a conservative lower bound of F along K , which defines by limit over comma

category [6, 5]:

$$(Ran_K F)(d) = \lim((d \downarrow K) \xrightarrow{\pi} C \xrightarrow{F} \mathcal{E})$$

The full formula state is to compute the value of $Ran_K F$ at some point d , take the limit over comma category. In the order of term by term,

- $(Ran_K F)(d)$ - evaluates extended functor F at new point d . Functor F only defines on known nodes (stated causal claims) (i.e. new point path "Sleep deprivation" $\rightarrow$ "hippocampus loss").
- $(d \downarrow K)$ - comma category, where all paths leads into d . For example, if $d =$ "hippocampus loss", then comma category contains things like cortisol $\rightarrow$ hippocampus loss (doc 2) and cortisol $\rightarrow$ hippocampus loss (doc 3). Gathering all evidence that points to d .
- $(\pi \rightarrow C \xrightarrow{F} \mathcal{E})$ - Once we collected all evidence in $(d \downarrow K)$, run them in this process where $\pi \rightarrow C$ strips the arrow and keeps the source (i.e. takes cortisol $\rightarrow$ hippocampus loss and throws away $\rightarrow$ hippocampus loss). The notation $\xrightarrow{F} \mathcal{E}$ has F as a lookup table from source nodes C to confidence scores. F looks up each nodes in the CSQL database and returns its score.
- $\lim(\dots)$ - merges all evidence conservatively. Since the limit (right Kan) is the intersection, it only accepts what all evidences agree on a constraint.

Essentially, the whole formula is read "to assign confidence to the novel claim at d , get every stated claim leading to d , look up their confidences under F , and merge them into the most conservative equation available.

Sheaves and Colimits: A sheaf on a site assigns local sections satisfying gluing axioms [1]. When two documents disagree on the direction of a causal edge (local sections conflict), they cannot glue globally. A sheaf requiring all papers to agree on the same site $\{A, B\}$ where paper 1 stated $B \rightarrow A$ and paper 2 stated $A \rightarrow B$ shows failure to the gluing axiom. Sheaf colimit resolves this by making it explicit that bidirectional relationship satisfies both: $A \leftrightarrow B$.

Topos of causal models: The CSQL database are modelled as presheaves on category document contexts [1]. The topos structure gives the classifier for claim filtering and category module for multi-document.

3 System Architecture

3.1 Workflow

After Democritus completes the retrieval and synthesis process, the newly created `democritus_discovery` file reads the CSQL database and apply two categorical concepts right Kan extension and Sheaf colimits using `run_causal_discovery.py`. This extension goes through the ordered process:

1. **CLIFF Querying:** The user submits their own query through CLIFF browser user interface. CLIFF routes it through the Democritus worker.
2. **Democritus:** Retrieves documents, extracts causal claims into `democritus_csql.sqlite`, which creates topos-based integrated HTML dashboard.
3. **Discovery program run:** The file `run_causal_discovery.py` finds the most recent CSQL database and applies a filter from the query tokens by running two category theory modules: Kan Extension and Sheaf Colimit.

4. **Kan Extension Module:** This constructs the category of stated claims and sets up the $A \rightarrow B \rightarrow C$ paths with shared bridge tokens. Then output the novel claims $A \rightarrow C$ using geometric mean confidence $(\text{Ran}_J F)(A \rightarrow C) = \sqrt{p \cdot q}$.
5. **Sheaf Colimit Module:** Verify for all claim pairs for any contradictions to claims such as $A \rightarrow B$ vs $B \rightarrow A$, check shared content tokens, and settle on the bidirectional feedback loop for those claims.
6. **Output:** JSON summary and a new HTML dashboard that depicts the claim relationships, bridge nodes, and confidence scores.

3.2 Domain Filtering

One of the challenges that this extension can face is that Democritus can drift off-topic depending on the user's input query, and without filtering, the discovery processes would chain edges of those unrelated documents together and produce incorrect information outputs. For example, using the sleep experiment research, one can ask the papers on sleep and cortisol. Democritus returns three papers, but one of the papers contain rheumatic disease paper either on autoimmunity or cartilage damage. That paper will get processed into the same CSQ database on our on-topic papers. This will create issues for Kan extension module in processing off-topic papers.

Before running any discovery files, the `democritus_discovery.py` file has code implementations of auto-detecting domain filter from the query:

1. Extract all content tokens of length ≥ 4 from user's query
2. Keep only causal edges whose normalized texts overlaps with at least one domain token.

You will see from later sleep deprivation query results that this would reduced around 401 edges to 177 domain-filtered edges. The domain filter also implements a safety feature where if fewer than 4 edges remain after filtering, the discovery file run prints a warning and stops rather than producing more results from irrelevant corpus.

4 The Kan Extension Module

4.1 How does it work?

The path build module finds every pair of causal edges that shared a bridge node:

$$\begin{aligned}
 &A \xrightarrow{\text{Relation}_1} B \text{ (from document } S_1, \text{ confidence } p) \\
 &B \xrightarrow{\text{Relation}_2} C \text{ (from document } S_2, \text{ confidence } q) \\
 &A \xrightarrow{\text{Relation}_1 + \text{Relation}_2} C \text{ (geometric-mean confidence } = \sqrt{p \cdot q})
 \end{aligned}$$

The **right Kan extension** of F along J is most conservative extension to all paths category [6, 5]. For a path of $A \rightarrow B \rightarrow C$ with confidence $p = \text{conf}(A \rightarrow B)$ and $q = \text{conf}(B \rightarrow C)$, which yields the geometric-mean confidence:

$$(\text{Ran}_J F)(A \rightarrow C) = \sqrt{\text{conf}(A \rightarrow B) \times \text{conf}(B \rightarrow C)} = \sqrt{p \cdot q}$$

When chaining two edges with confidence p and q , we use the geometric mean $\text{conf}(A \rightarrow C) = \sqrt{p \cdot q}$ [4]. We use geometric mean over arithmetic mean to avoid over-confidence for claims with weak support. For example, if $p = 1.0$ and $q = 0.01$, the arithmetic mean gives 0.505 which may be too confident. It is also that the geometric mean is $\leq$ arithmetic mean, making it a conservative choice.

4.2 Semantic Filter

Path composition can create misleading chaining depending on how the two edges chain together. Two edges can share an intermediate node label without actually referring to the same concept. To avoid this, it is required that the bridge node B shares at least $\text{MIN_OVERLAP} = 2$ content tokens in common with both the $A \rightarrow B$ and $B \rightarrow C$ edges.

The reason for two tokens is because a single token match (i.e. both edges mention "risk") could happen in coincidence or by accident. Two shared tokens would force a bridge to have semantic content on both sides. This requirement implements a condition that creates path forming a valid pathing that is not accidental.

4.3 Examples of Claims from experiment

This table depicts Kan Extension claims produced on sleep deprivation corpus. None of the documents stated any of such $A \rightarrow C$ edges explicitly.

Table 1: Kan Extension claims (sleep deprivation corpus, $n = 3$)

Conf.	Source A	Relation	Target C
0.33	childhood obesity increases insulin resistance	causes	higher blood pressure
0.33	obesity increases insulin resistance	causes	higher blood pressure
0.33	childhood obesity	causes	higher blood pressure
0.33	chronic inflammation	increases	risk of cardiovascular complications

5 The Sheaf Colimit Module

5.1 How does it work?

The conflict resolution module scans all pairs of domain-filtered edges for direction conflicts:

1. Edge 1: $A \rightarrow B$ (from document S_1)
2. Edge 2: $B \rightarrow A$ (from document S_2)

With S_1 asserting local section $A \rightarrow B$ over $\{A, B\}$ and S_2 asserting conflicting local section $B \rightarrow A$ over the same site, it violates the gluing axiom as it [1]. This Sheaf Colimit module resolves this through bidirectional feedback loop $A \leftrightarrow B$.

Here, the harmonic-mean confidence of the feedback loop is:

$$\text{conf}(A \leftrightarrow B) = \frac{2pq}{p+q}, p = \text{conf}(A \rightarrow B), q = \text{conf}(B \rightarrow A)$$

- Harmonic mean - more conservative than geometric mean when p and q are far apart
- For conflicting evidence - if one leg of loop is weak, the whole loop should be penalized
- Geometric mean already used in Kan Extension for chaining

5.2 Example of Feedback Loop experiment

Table 2 shows top Sheaf Colimit claims. None were stated bidirectionally in any source document:

Table 2: Get sheaf colimit feedback loops (sleep deprivation corpus)

Conf.	Feedback Loop $A \leftrightarrow B$
0.44	sleep deprivation $\leftrightarrow$ cognitive performance in adults
0.33	apnea induced intermittent hypoxia increases oxidative stress which $\leftrightarrow$ endothelial dysfunction and contributes to cardiovascular disease
0.33	autoimmunity increases inflammation which $\leftrightarrow$ progression of joint damage in rheumatic diseases
0.33	change in the mediator variable $\leftrightarrow$ total effect of the treatment on the final outcome

The top loop for sleep deprivation $\leftrightarrow$ cognitive performance with $\text{conf} = 0.44$ was supported by two documents on $A \rightarrow B$ side $p = 0.67$ and one on $B \rightarrow A$ side $q = 0.33$, which gives a mean of $2 \times 0.67 \times 0.33 / (0.67 + 0.33) = 0.44$.

6 Implementation

6.1 Code Structure

Two of the newly implemented files were added in the `functorflow_v3` package:

- `democritus_discovery.py` - main categorical modules (Kan extension and Sheaf colimits), SQLite reader and JSON/HTML output
- `run_causal_discovery.py` - high level processor that locates CSQl database and readers run metadata, starts democritus discovery, and opens browser

There were several tweaks on other existing files to accommodate the newly created files `democritus_discovery.py` and `run_causal_discovery.py` such as: `query_router_agent.py`, `democritus_batch_agentic.py`, `democritus_agentic.py`, `cliff_worker.py`.

6.2 Running the System

```
# Step 1: Run CLIFF and query
python -m functorflow_v3.cliff --outdir ./cliff_results_v2 '
--course-repo-root "C:\Users\hache\Documents\GitHub\Category-Theory-for-AGI-UMass-
CMPSCI-692CT"
# Then submit user query in the browser

# Step 2: Run categorical discovery on sqlite produced. Run with explicit constraints:
python -m functorflow_v3.democritus_discovery '
```

```

--outdir .\cliff_results_v2 '
--query "Find me 10 studies in the bidirectional relationship between sleep deprivation
and cortisol, including papers that disagree on the primary causal direction." '
--min-confidence 0.10

# Or run complete discovery (locates most recent CSQL)
python functorflow_v3/run_causal_discovery.py \
--skip-cliff --min-confidence 0.10

```

Listing 1: Complete run commands for this extension

6.3 Outputted Validation

What is outputted from `_validate_result` function in `run_causal_discovery.py` prints out warnings when:

- no documents retrieved or empty corpus
- fewer than `MIN_DOCS_FOR_SHEAF` = 2 documents were retrieved, where sheaf colimit cannot find cross-document conflicts
- fewer than `MIN_EDGES_FOR_KAN` = 4 domain-filtered edges remain, where Kan extension could not produce a path.

These guardrails will address the issues when a certain user query cause Democritus to wander off-topic, deterring the process instead of creating misleading novel claims from irrelevant corpus.

7 Experiments and Quantitative Results

7.1 Corpus and Query

The main evaluation of a corpus was produced by a recent CLIFF query:

“Find me 10 studies in the bidirectional relationship between sleep deprivation and cortisol, including papers that disagree on the primary causal direction.”

Democritus retrieves 3 documents stored in a CSQL database, though the query requested up to 10 studies. This reflects on the combination of public availability constraints, query specifics, and early stopping once the retrieved corpus stabilizes.

Topos Synthesis dashboard indentifies two topics: **Sleep** (2 documents; main topics on sleep, data, thyroid) and **Rheumatic** (1 document; main topics: rheumatic, disease, snps impacts). The Rheumatic topic was off-topic, which is an issue to be addressed by weakening through domain filter.

7.2 Additional Experiments

Three other queries were submitted during this experiment:

- **GL-1 receptor agonists** (3 documents): original domain. Democritus correctly identified GLP-1 drugs, obesity, insulin resistance, and cardiovascular risk.
- **Depression causing cardiovascular disease** (3 documents): linked chronic stress, cortisol dysregulation, and cardiac risk.

Table 3: Discovery statistics on sleep deprivation corpus

Info Metric	Value
Numer of Source documents	3
Raw causal edges before domain filter applied	401
After Domain-filtered edges	177
Connecting nodes with ≥ 1 successor	83/177 (47%)
Kan Extension Claims (novel)	40
Polarity conflicts found	31
Sheaf Colimit Feedback Loop (novel)	20
Total Novel claims	60

- **Sleep characteristics and thyroid function:** consistent with bidirectional sleep corpus, which confirms domain filter used across query.

8 Discussion

8.1 Novel Claims and Source Documents

Most of the Kan Extension claims in the sleep corpus experiment connects back to childhood obesity through insulin resistance to elevated blood pressure and sleep apnea. Such chains can be scientifically well-supported as the nodes correctly bridge them. The 0.33 confidence reflects on single-document support, given small corpus.

The top Sheaf Colimit claim (sleep deprivation $\leftrightarrow$ cognitive performance $\text{conf} = 0.44$) is also well-supported: the bidirectional relationship between sleep loss and cognitive decline is supported in the documents, with results from sleep disturbance through rumination. The modules correctly identifies this as feedback loop without any single source paper stating it bidirectionally.

8.2 Limitations

- **Small corpus:** With only 3 documents, the sheaf colimit module has limited cross-document conflicts. A larger corpus would produce denser conflict structure and higher confidence.
- **Topic drift:** Thhe rheumatic partition example shows that Democritus can sometimes retrieve off-topic documents. Could create a flag that can provide stricter containment.
- **No logical closings:** Since the Kan Extension module implements only 1 depth path $A \rightarrow B \rightarrow C$, it may seem shallow. An iterated application with a deeper path would require some fixed iteration to avoid exponential run.
- **Confidence calibration:** The geometric mean is a valid approach but only approximate. There are other methods that can be more accurate over the CSQL graph, especially through longer chains and accounting for shared evidence.

9 Conclusion

What is presented is a categorical causal discovery layer for CLIFF AGI system [2]. The layer has two modules relating to course materials: the Kan Extension [5, 6] for constructing and discovering novel causal chains by stated edges and the Sheaf Colimit [1] in discovering bidirectional feedback loops from contradictory local claims. Applied to the corpus is the three documents on sleep

deprivations. The system produces 60 novel claims, 40 paths, and 20 bidirectional feedback loops, which appears either none or implicitly in the source documents.

The project extension shows that category theory can be used to generate outputs in different possibilities through AI systems. Kan extension and sheaf colimits can be used as algorithms where CLIFF system can integrate with using CSQL databases and domain filtering edges. Such algorithms can be run through real research corpuses.

The implementations shows every novel claim derived directly from the retrieved corpus. Every edge in the Kan Extension or Sheaf Colimit outputs can be traced back to at least two stated edges with document IDs and domain filters to make sure the synthesis stays on-topic. This categorical idea allows the system to find truth within the gaps between documents but also stays on track to the context of the documents.

References

- [1] S. Mahadevan, *Categories for AGI*. University of Massachusetts Amherst / Adobe Research, 2026. Available at: <https://people.cs.umass.edu/~mahadeva/papers/catagi.pdf>
- [2] S. Mahadevan, *CLIFF_CatAgi: CLIFF is the first AGI chatbot designed with Functor Flow technology*. GitHub repository, 2026. https://github.com/sridharmahadevan/CLIFF_CatAgi
- [3] S. Mahadevan, *AGI and Consciousness*. Lecture slides, CMPSCI 692CT, University of Massachusetts Amherst, 2026.
- [4] S. Mahadevan, *Kan Extension and TopoCoend Transformers: Categorical Attention Models*. Lecture slides, CMPSCI 692CT, University of Massachusetts Amherst, 2026.
- [5] S. Mahadevan, *Universal Decisions with Kan Extensions*. Lecture slides, CMPSCI 692CT, University of Massachusetts Amherst, 2026. See also: *Universal Decision Model*, arXiv:2110.15431, 2021.
- [6] E. Riehl, *Category Theory in Context*. Dover Publications, 2016. Available at: <https://emilyriehl.github.io/files/context.pdf>
- [7] B. J. Baars, *A Cognitive Theory of Consciousness*. Cambridge University Press, 1988.